SOUMIKA DAS

Software Engineer | Full-Stack & Backend Development | AI/ML

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Professional Summary

Computer Science undergraduate and Software Engineer with four internships across FinTech, HealthTech and research, building full-stack platforms, real-time systems and AI services. Published deep learning researcher and GDG speaker.

Education

B.Tech in Computer Science and Engineering (IoT, Cybersecurity and Blockchain Technology) Expected 2027
Institute of Engineering and Management, Kolkata CGPA: 7.69/10

Core Skills

- **Languages:** Python, JavaScript, TypeScript, Java, C++, SQL
- **Frontend:** React, Next.js, HTML, CSS, Tailwind CSS, Vite
- **Backend:** Node.js, Express, FastAPI, Flask, Django, REST APIs, WebSocket, Microservices, JWT/OAuth
- **Databases:** PostgreSQL, MySQL, MS SQL Server, MongoDB, Vector DB
- **Cloud & DevOps:** Google Cloud (GCP), Vercel, Render, Docker, Linux, Nginx, CI/CD, Git/GitHub, Ansible
- **AI/ML:** GenAI, LLMs, RAG, Agentic AI, PyTorch, TensorFlow, OpenCV, NLP, CNN, LSTM, YOLO
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Unit Testing

Projects

BONAP - Botanical Scraper & Plant Portal | Live App

- Developed a Node.js scraper and ETL pipeline pulling taxonomy, distribution maps and morphology from BONAP, USDA and GBIF into a Vite + React comparison portal covering **26K+ species** across 4K+ genera, with AI summaries.

Brain Tumor Classification - Ensemble Deep Learning

- Engineered a CNN-LSTM ensemble classifying brain tumors from **3,200+ MRI scans**, reducing single-model variance.

DeepTrust - Degree Verification API | GitHub

- Designed a Flask API verifying degrees via QR codes and OpenCV/Tesseract OCR, supporting PDFs and **6 image formats**.

Internship Experience

PepHub - Software Engineer Intern (Oct 2025 - Present) | Remote

- Delivered multiple client projects, including a Next.js/Node.js portal with **Stripe payments** and a FinTech REST API.
- Automated production releases with PM2 and SSH, cutting deployment time by **82%** (45 min to 8 min).

IEDC, Dept. of CSE, UEM Kolkata - Research Engineer Intern (Dec 2024 - Dec 2025) | Hybrid

- Reduced HPC cluster setup **95%** (8 hrs to 22 min) via Ansible; built YOLO/OpenCV face detection at **96%** accuracy.

Cardiacrhythm - Software Engineer Intern (Jun 2025 - Sep 2025) | Remote

- Architected WebSocket real-time ECG streaming for **200+ concurrent streams**, reducing latency by **90%** (1.8s to 180ms); shrank payloads **75%** (8MB to 2MB); added auto-reconnect, preventing data loss in **15K+ daily sessions**.

RevMaxx - Software Engineer Intern (Jan 2025 - Jun 2025) | Onsite

- Built a **HIPAA-compliant** transcription backend with SNOMED CT/ICD-10 coding and an NLP entity-extraction service across **50K+ clinical notes**; added retry logic and circuit breakers, cutting failures by **over 90%** (12/week to under 1).

Publications & Leadership

- **Enhancing Brain Tumor Diagnosis with Ensemble Deep Learning Technique for Improved Accuracy**, AISC 2025 (Springer) | LinkedIn
- **Cloud Lead & Speaker**, GDG on Campus UEM Kolkata (2025 - 26): led the Cloud domain, conducted Google Cloud Study Jams, spoke at “Tech and Adda” and co-organized the “StackUp!!” workshop. | LinkedIn